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B.Tech. Degree VI Semester Examination April 2019

ME 15-1604 HEAT AND MASS TRANSFER (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) Derive the equation for conduction of heat through a slab or plane wall.
- (b) What do you mean by extended surfaces and state their applications.
- (c) Write a short note on lumped heat capacity systems.
- (d) Derive energy equation for thermal boundary layer over a flat plate.
- (e) State and explain Ficks law of diffusion.
- (f) Differentiate between the mechanism of filmwise and dropwise condensation.
- (g) State and prove Kirchhoff's law of radiation.
- (h) Derive an expression for shape factor in the case of radiation exchange between two surfaces.
- (i) Explain various types of heat exchangers with neat sketches.
- (j) What do you mean by 'fouling' in heat exchangers?

PART B

(4 × 10 = 40)

- II. (a) Derive general heat conduction equation in cylindrical coordinates. (5)
 - (b) A spherical container of negligible thickness holding a hot fluid at 140°C and having an outer diameter of 0.4 m is insulated with three layers of each 50 mm thick insulation of $k_1 = 0.02$ W/mK, $k_2 = 0.06$ W/mK and $k_3 = 0.16$ W/mK (starting from inside). The outside surface temperature is 30°C. Determine (i) the heat loss (ii) interface temperature of insulating layers. (5)
- OR**
- III. (a) A motor body is 360 mm in diameter (outside) and 240 mm long. Its surface temperature should not exceed 55°C when dissipating 340 W. Longitudinal fins of 15 mm thickness and 40 mm height are proposed. The convection coefficient is 40 W/m².°C. Determine the number of fins required. Atmospheric temperature is 30°C, thermal conductivity = 40 W/m°C. (5)
 - (b) Derive an expression for critical radius of insulation for a cylinder. (5)
- IV. (a) Air at 20°C at atmospheric pressure flows over a flat plate at a velocity of 3 m/s. If the plate is 1 m wide and 80°C, calculate the following at $x = 300$ mm. (7)
 - (i) Hydrodynamic boundary layer thickness
 - (ii) Thermal boundary layer thickness
 - (iii) Local friction coefficient
 - (iv) Average friction coefficient
 - (v) Local heat transfer coefficient
 - (vi) Average heat transfer coefficient
 - (vii) Heat transfer
 - (b) A vertical pipe 80 mm diameter and 2 m height is maintained at a constant temperature of 120°C. The pipe is surrounded by still atmospheric air at 30°C. Find heat loss by natural convection. (3)

OR

(P.T.O.)

- V. (a) Water is boiled at the rate of 24 kg/hr in a polished copper pan, 300 mm in diameter, at atmospheric pressure. Assuming nucleate boiling conditions, calculate the temperature of the bottom surface of the pan. (5)
- (b) An open pan 20 cm in diameter and 8 cm deep contains water at 25°C and is exposed to dry atmospheric air. If the rate of diffusion of water vapour is 8.54×10^{-4} kg/hr, estimate the diffusion co-efficient of water in air. (5)
- VI. (a) The sun emits maximum radiation at $\lambda = 0.52 \mu$. Assuming the sun to be a black body, calculate the surface temperature of the sun. Also calculate the monochromatic emissive power of the sun's surface. (4)
- (b) Two large plates are maintained at a temperature of 900 K and 500 K respectively. Each plate has area of 6 m². Compare the net heat exchange between the plates for the following cases. (6)
- (i) Both plates are black.
- (ii) Plates have an emissivity of 0.5.
- OR**
- VII. (a) Two parallel plates of size 1.0 m by 1.0 m spaced 0.5 m apart are located in a very large room, the walls of which are maintained at a temperature of 27°C. One plate is maintained at a temperature of 900°C and other at 400°C. Their emissivities are 0.2 and 0.5 respectively. If the plates exchange heat between themselves and the surrounding, find the net heat transfer to each plate and to the room. Consider only the plate surface facing each other. (5)
- (b) Calculate the net radiant heat exchange per m² area for two large parallel plates at temperatures of 427°C and 27°C. ϵ (hot plate) = 0.9 and ϵ (cold plate) = 0.6. If a polished aluminium shield is placed between them, find the % reduction in the heat transfer. ϵ (shield) = 0.4. (5)
- VIII. (a) A counter flow heat exchanger is employed to cool 0.55 kg/s ($C_p = 2.45$ kJ/kg°C) of oil from 115°C to 40°C by the use of water. The inlet and outlet temperature of cooling water are 15°C and 75°C respectively. The overall heat transfer coefficient is expected to be 1450 W/m²K. (6)
- Using NTU method, calculate the following:
- (i) The mass flow rate of water
- (ii) The effectiveness of heat exchanger
- (iii) The surface area required
- (b) Water flows at the rate of 65 kg/min through a double pipe counter flow heat exchanger. Water is heated from 50°C to 75°C by an oil flowing through the tube. The specific heat of the oil is 1.780 kJ/kg.K. The oil enters at 115°C and leaves at 70°C. The overall heat transfer co-efficient is 340 W/m²K. Calculate the following: (4)
- (i) Heat exchanger area
- (ii) Rate of heat transfer
- OR**
- IX. (a) A heat exchanger is to be designed to condense an organic vapour at a rate of 500 kg/min which is available at its saturation temperature of 355 K. Cooling water at 286 K is available at a flow rate of 60 kg/s. The overall heat transfer coefficient is 475 W/m²K. Latent heat of condensation of the organic vapour is 600 kJ/kg. Calculate: (6)
- (i) The number of tubes required, if tubes of 25 mm outer diameter, 2 mm thick and 4.87 m long are available
- (ii) The number of tube passes, if cooling water velocity (tube side) should not exceed 2 m/s
- (b) Derive the effectiveness of a counter-flow heat exchanger. (4)